NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 06

Implementation Plan

Completed work with evidence, and remaining work ordered by dependency with priorities, estimates, acceptance criteria and tests.

Document control

Document ID	NX-06-PLAN
Version	1.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Development team planning the final PoE; assessor reviewing project management
Evidence baseline	Branch New-Changes at commit 3a3f275 (tree identical to main at 9f8e82a)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

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1 Completed work

This history is reconstructed from `PROJECT_IMPROVEMENTS.md`, merged pull requests and commit messages. Dates are commit or merge dates **Implemented**.

Delivered increments

Date	Increment	Evidence	Requirements
2026-08-23	Initial Android app with auth integration, secure token storage, nav flow, splash animation	Commits 65385b2, 0496717, 964cbf5	FR-01, FR-02, FR-19
2026-09-22	CI workflow, Room read cache, task due date / priority / status / labels / checklist, dashboard insights, auth validation	PRs #10–#16; commit 8b31aa3	FR-09, FR-11
2026-09-23	Backend remediation: mutex + atomic writes, Zod validation, JWT secret check, bcrypt, rate limiting, health check, Pino; task detail screen	PR #23; `server.test.js`	NFR-01, NFR-04, FR-12
2026-09-24	Android structural remediation: six repositories, manual DI, `HabitScoring` domain, feature packages, type-safe routes, shared fakes	PR #24	TR-01 to TR-03, FR-14
2026-09-26	UI refinement: theme colours, logo component	PR #25	NX-04-DES
2026-10-04	Offline upload queue and `/api/sync`, sync banner, conflict review, project search, task filters, zu/tn resources, configurable `NEXUS_API_URL`, HTTPS-only release, Dockerfile/compose	PR #26 (commit 3a3f275)	FR-16, FR-17, FR-18, TR-07

2 Remaining work

Tasks are ordered so that each depends only on tasks above it. Priority: **P0** blocks a credible release or submission, **P1** is needed before real users, **P2** is improvement. All effort figures are **Estimate** person-day estimates by the document author, not team commitments, and should be re-estimated by the team.

Dependency-ordered backlog

ID	Task	Pri.	Depends on	Est. (days)	Acceptance criteria	Tests
IMP-01	Restore GitHub Actions execution (resolve the account billing lock shown in run annotations) and re-run CI on `main`	P0	None	0.25 **Estimate**	Android CI and Backend CI both green on `main` at or after 3a3f275	Existing CI jobs (CI-01, CI-02)
IMP-02	Fix Zod 4 compatibility before merging PR #27: use `error.issues`; give `z.record` a key schema	P0	IMP-01	0.5 **Estimate**	Backend tests pass on Zod 4; validation still returns 400 `VALIDATION_ERROR`	`npm test`; existing Zod tests

ID	Task	Pri.	Depends on	Est. (days)	Acceptance criteria	Tests
IMP-03	Fail safe on an unreadable data file: refuse to start (or start read-only) instead of starting empty	P0	None	0.5 Estimate	A corrupt `nexus-data.json` stops the process with a clear log line and leaves the file untouched	New Jest test with a corrupt fixture (AUD-01)
IMP-04	Correct the auth error messages: 401 at login gives "Email or password is incorrect"; 409 gives "Email already registered"; 429 gives "Too many attempts"	P0	None	0.5 Estimate	Each status shows its own message; session expiry still shows the expiry text elsewhere	`AuthViewModelTest` cases per status (AUD-18)
IMP-05	Remove or fix the root `docker-compose.yml`; document `backend/compose.yaml` as the single container path	P1	None	0.25 Estimate	`docker compose up` from the documented path persists data across restarts	Backend CI smoke test plus a manual restart check (AUD-02)
IMP-06	Add `app.set("trust proxy", 1)` (configurable) for TLS-proxy hosting	P1	IMP-05	0.25 Estimate	The rate limit keys on the client IP behind the chosen proxy	Supertest with `X-Forwarded-For` (AUD-06)
IMP-07	Provision the hosted API and release signing; set the `NEXUS_API_URL` repository variable	P1	IMP-01, IMP-03, IMP-05, IMP-06	1–2 Estimate	`/health` reachable over HTTPS; signed release APK/AAB produced by CI	Release checklist in NX-07-CICD (AUD-08)
IMP-08	Due-date reminders, or hide the switch (decision OQ-01)	P1	OQ-01	0.25 (hide) / 3–5 (build) Estimate	The switch either has a visible effect or is not shown	Unit test on the scheduler, or UI test that the switch is absent (AUD-10)
IMP-09	Exclude the Room database from backup, or document the choice	P1	None	0.25 Estimate	`backup_rules.xml` and `data_extraction_rules.xml` exclude `nexus_database`	Manual `adb shell bmgr` check (AUD-05)
IMP-10	Enforce lint in CI (`ktlintCheck`, `detekt`) and fix the ktlint parse failure; delete the placeholder `blank.yml` and `build-verification.log`	P1	IMP-01	1 Estimate	CI fails on lint violations; the log file is no longer tracked	CI run (AUD-13)
IMP-11	Accessibility pass: TalkBack walkthrough, contrast measurement, 48 dp targets	P1	None	1–2 Estimate	Results recorded; issues fixed or logged	Accessibility Scanner report; Compose semantics tests

ID	Task	Pri.	Depends on	Est. (days)	Acceptance criteria	Tests
IMP-12	Undo for list deletes of projects, tasks and habits	P2	None	1 Estimate	Snackbar "Undo" restores the item before upload	ViewModel tests
IMP-13	CSV export escaping (RFC 4180 quoting)	P2	None	0.25 Estimate	A habit named `Read, write` exports as one quoted field	`SettingsViewModelTest` (AUD-11)
IMP-14	Fix the habit completion-rate formula: count habit-date pairs	P2	None	0.25 Estimate	Two daily habits both done for 30 days gives 100%	`HabitScoringTest` (AUD-17)
IMP-15	Enable Room `exportSchema` and add `MigrationTestHelper` tests for 3→4→5	P2	None	1 Estimate	Schema JSON committed; migration tests pass	Instrumented tests (AUD-12)
IMP-16	Documentation corrections: README contents links, admin role label, translation claim, recaptured screenshots	P2	IMP-07	0.5 Estimate	Every README link resolves; screenshots match their file names	Link check (AUD-09, AUD-21)
IMP-17	Complete feature-screen localisation (zu/tn)	P2	None	3–5 Estimate	No hard-coded user-facing English strings remain in `ui/`	Lint `HardcodedText`; pseudo-locale run (FR-18)
IMP-18	Final-PoE features (Google SSO, push notifications), as decided in OQ-04	P2	IMP-07, OQ-04	Unknown	Defined per feature in `docs/POE_PART_2_ADDENDUM.md`	Server and end-to-end tests required by the addendum

Total for P0 and P1 excluding IMP-08 "build" and IMP-18: about 5–7 person-days Estimate.

3 Definition of done

Adopted from the quality checklist in `PROJECT_IMPROVEMENTS.md`: API models and validation first; a Room migration for any schema change; loading, offline, retry, unauthorised and validation states covered; ViewModel unit tests plus instrumented coverage of the main path; accessibility checked; README and the improvements log updated Implemented. Recommendation Add "CI green on the PR" once IMP-01 is done.